The Prevalence of Anxiety and Depression Disorders with Demographic and Socio-Economic Factors : the Case of the United States

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Abstract

Mental health issues, particularly anxiety and depression, have become a major concern for governments, as the costs associated with lost productivity are significant. However, the socio-economic factors contributing to their prevalence remain widely debated. This study challenges the common belief that people from disadvantaged socio-economic backgrounds are the most affected. On the contrary, it reveals that people with higher levels of education and more demanding jobs face greater mental health problems. By analyzing a wide range of demographic and socio-economic variables, the research highlights the hidden costs of work stress and academic pressure, underlining the urgent need for proactive mental health policies.

1 Introduction

Health is a fundamental component of a country's development, influencing productivity, economic growth and social well-being, as demonstrated by Bloom and Canning (2000). This study focuses on one area of health : mental health and specifically on anxiety and depression disorders. The National Alliance of Mental Illnesses (NAMI) states that by 2021, 22.8% of the U.S. population - approximately 1 in 5 adults - suffered from a mental illness[1]. The situation has worsened with the COVID-19 pandemic, indeed according to the Organisation for Economic Co-operation and Development (OECD), the proportion of people reporting symptoms of depression and anxiety has doubled across all OECD countries, with a particularly significant increase in the United States[2]. Indeed in 2019, the proportion of people experiencing depression was around 6.5%, rising sharply to 22.9% in 2020[2].

Mental health is a major health problem that strongly affects individuals and the overall economy. Besides, Knapp and Wong (2020) assert that mental illnesses lead to difficulties in completing education, meeting professional requirements and contributing to the social and economic marginalization of individuals. Trautmann et al. (2016) underlined that mental illness implies not only "visible costs" such as diagnosis and treatment in healthcare systems costs but also "invisible costs" related to income losses caused by a loss of productivity due to the disability. For this purpose, World Health Organisation (WHO) in 2022 estimates that 12 billion productive work days are lost due to depression and anxiety disorders alone with a cost for the global economy of US $1 trillion a year in lost productivity[3]. Understanding the determinants of anxiety and depression is not only essential for improving individual well-being but also for mitigating the economic and social costs associated with these conditions. In this context, the aim of this study is to understand how demographic and socio-economic factors could impact the prevalence of anxiety and depression in the United States.

Firstly according to the literature, Naima Z. Farhane-Medina (2022) found that women tend to be more affected by anxiety disorders than men. A possible explanation lies in the brain regions involved in emotion regulation, such as amygdala, which responds differently to emotions between men and women. Similarly, Lowe et al. (2009) show that women are, on

[1]National Alliance on Mental Illness (2023), "Mental Health By the Numbers"
[2]OECD Mental health
[3]World Health Organization. (2024). Mental health at work.

average, more affected also by depression than men. This divergence is explained by hormonal variations, women's greater concern with body image and a tendency toward more negative attributional styles. Overall, a similar trend emerges for both anxiety and depression from the existing literature. The level of education has also an impact on anxiety and depression. In fact, Kondirolli and Sunder (2022) found that as the level of education increases, the probability of suffering from anxiety and depression decreases, by demonstrating that : *"An additional year of education led to a lower likelihood of reporting any symptoms related to depression (11.3%) and anxiety (9.8%)"*. In addition, Joannès et al. (2023) established a negative relationship between education and anxiety explained by the fact that individuals with lower levels of education are more affected by anxiety due to economic stress related to the difficulty in entering a competitive labor market. Age is another significant factor in explaining mental illnesses. Westerhof and Keyes (2010), through an empirical analysis of data on individuals aged 18 to 65 years and older, found that older adults generally experience fewer mental health issues, except for those in the most advanced age group. However, a distinction between depression and anxiety should be noted, as Zenebe et al. (2021) estimated the average prevalence of depression among the elderly population to be no less than 31.74%, highlighting the substantial impact of age on depression. Conversely, younger individuals appear more vulnerable to anxiety disorders. Goodwin et al. (2020) found that anxiety levels among the US population have increased, particularly among young adults aged 18 to 25 years. One possible explanation for this growing prevalence is the rise of social media, which has been linked to increased anxiety among young people. In particular, Paul and Moser (2009) have noticed that people that are unemployed are more likely to suffer from mental illnesses such as anxiety, depression and symptoms of distress. This association may stem from factors such as financial stress, loss of routine and reduced social interactions. According to the study by Palner and Mittelmark (2002), mental distress is also affected by relational factors, such as marital status. Indeed, separated and divorced individuals are more affected by mental health issues than married people. This finding aligns with the research of Walen and Lachman (2000) suggesting that social interactions with one's partner play a consistently positive role in predicting well-being and health. While support from family and friends also contributes to mental well-being, its impact is smaller compared to that of a partner's support. The race of an individual has also a role to play. Indeed, the "Black-White Mental Health Paradox" explained by Erving et al. (2019) refers to the phenomenon in which black individuals in the United States tend to report lower rates of depression and anxiety than white individuals, despite facing greater exposure to stressors such as racism, poverty, and discrimination that can negatively affect mental health. Research suggests that this paradox may be explained by cultural differences in coping strategies, strong support systems within Black communities, and how mental health symptoms are understood and reported. Another determinant has to be considered, Moschion and van Ours (2022) found that depression increases the risk of homelessness especially among men and experiencing homelessness further raises anxiety disorder risk. These findings suggest that unstable housing creates a feedback loop that worsens mental health outcome.

In short, the literature has shown that demographic and socio-economic factors such as gender, education, age, employment status, living environment and marital status play a role in the prevalence of anxiety and depression. However, existing research has not fully examined the combined effect of several socio-economic determinants. This study aims to fill these gaps by analyzing a large sample of data of 15 millions of observations in the Unites States collected between 2018 and 2022, allowing a more complete perspective on the factors affecting the prevalence of anxiety and depression. The empirical analysis relies on various econometric models, including simple and interaction models, selected based on their relevance to addressing the research question. We will use the probit model and robust standard errors to estimate the econometric models. Our results indicate that individuals who are women, reside in the Midwest, possess a high level of education, and hold a full-time job contract have a higher likelihood of experiencing anxiety and depression. However, the determinants of anxiety and depression can differ. Anxiety is more prevalent among younger age groups, whereas depression affects both younger and older individuals.

The first part of this paper outlines the study's context, followed by a description of the dataset used for the empirical analysis. It then presents the econometric models selected for the study and the final section focuses on analyzing the results and drawing conclusions.

2 Institutional Context and Administrative Data

2.1 Institutional Context

Anxiety disorder is defined by WHO[4] as an *"excessive fear or worry about a specific situation or, in the case of generalized anxiety disorder, about a broad range of everyday situations"*. These feelings are often accompanied by physical tension and cognitive symptoms such as difficulty sleeping, trouble making decisions and nausea. According to NAMI[5] anxiety disorders are the most prevalent mental health condition in the United States, affecting over 40 million adults or 19.1% of the population. The prevalence of anxiety symptoms has increased significantly in recent years. The percentage of adults experiencing anxiety symptoms rose from 15.6% in 2019 to 18.2% in 2022 demonstrated by Terlizzi and Zablotsky (2024).

Depression disorder is characterized by WHO[6] as *"a low mood or loss of pleasure or interest in activities for long periods of time"*. The symptoms related could be sleep troubles, disturbances in appetite, feeling of low-self worth or thought of dying[6]. People who have experienced abuse or severe losses during their lives are more likely to suffer from depression[6]. Depression is considered as a common mental health disorder, in 2020 18.4% of U.S. adults reported having ever been diagnosed with depression by Lee et al. (2023). Depression symptoms in the U.S. notably increased, Terlizzi and Zablotsky (2024) indicates that the percentage of adults experiencing any symptoms of depression grew from 18.5% in 2019 to 21.4% in 2022. Over the past decades in the U.S., legislative efforts have aimed to improve access to mental health care. Notably, the Mental Health Parity Act (MHPA) and the Mental Health Parity and Addiction Equity Act (MHPAEA) are federal laws that ensure mental health and substance use disorder treatments receive the same level of coverage as medical and surgical care[7]. In 2010, the Affordable Care Act (ACA) further strengthened parity laws by mandating that most health plans provide coverage for mental health and substance use disorder care[7].

Despite the high prevalence of anxiety and depressive disorders in the United States and policy reforms, access to appropriate treatment remains a significant challenge. According

[4] World Health Organization (2023). Anxiety disorders.
[5] National Alliance on Mental Illness (NAMI). (2017). Anxiety disorders.
[6] World Health Organization. (n.d.). Depression
[7] National Alliance on Mental Illness (NAMI). (n.d.). Mental health parity.

to Young et al. (2001), while 83% of American adults with these conditions have visited a healthcare provider, only 30% receive treatment that aligns with clinical guidelines. Examining socioeconomic influences on anxiety and depression reveals also key insights into ongoing challenges in U.S. mental health care access and equity.

2.2 Administrative Data

This empirical research relies on the Mental Health Client Level Data (MH-CLD), provided by a U.S. government agency called Substance Abuse and Mental Health Services Administration (SAMHSA). The MH-CLD dataset consists of detailed annual records of 6 to 7 million individuals who accessed mental health services in the United States between 2018 and 2022. These records are collected through state administrative data systems that track clients receiving mental health treatment. The dataset includes a wide range of socio-demographic and clinical variables, which allow a deep analysis of the research question. To facilitate a over-time analysis, these records were combined into a pooled cross-sectional dataset. However due to the size of the combined dataset, missing values (NA) in the variable 'education' were removed from each dataset to be able to exploit the whole dataset. This preprocessing step resulted in a large-scale dataset of 15,605,775 observations, allowing for an in-depth analysis of the socioeconomic determinants of anxiety and depression.

Finally, the variables chosen to address this research question are detailed in Table 1 of the Appendix. The paper examines how these variables affect anxiety and depression prevalence in the U.S., providing insights for policy and intervention strategies.

3 Model and Estimation

3.1 Econometrics Models

3.1.1 Simple Models

The estimated econometric model is :

$$Y_i = \beta_0 + \sum_{j=1}^{n} \beta_j X_j + \varepsilon_i \tag{1}$$

where Y_i represents Depression or Anxiety, and X_j includes Gender, Age, Education level, Residential Region, Race, Residential Status, Not in Labor Force Status, and Marital Status (see details in section A.1). β_0 is the intercept, β_j measures the marginal effect of X_j on X_i, and ε_i is the error term.

The results of these regressions are shown in Tables 2 and 5. While, the marginal effects at means for these models are presented in Tables 3 and 6.

3.1.2 Models with interactions

Models with interaction terms capture the impact of employment status on disorder prevalence while incorporating relevant identified interactions.

The following econometric models are estimated :

$$\text{Anxiety} = \beta_0 + \sum_{i=1}^{n} \beta_i X_i + \varepsilon \tag{2}$$

X_i: Employment Status, Employment Status * Education, Employment Status * Region, Employment Status * Gender and Employment Status * Age

The results of the regression are shown in Table 4.

$$\text{Depression} = \beta_0 + \sum_{i=1}^{n} \beta_i X_i + \varepsilon \tag{3}$$

X_i: Employment Status, Employment Status * Education, Employment Status * Region and Employment Status * Gender

The results of the regression are shown in Table 7.

3.2 Estimation Method

A probit model is used to analyze the determinants of anxiety and depression diagnoses. This econometric approach is particularly suited for binary dependent variables, where the outcome is coded as 1 if the individual has been diagnosed with anxiety or depression and 0 otherwise. The probit model assumes the existence of an underlying latent variable that represents the probability of experiencing anxiety or depression. This probability is influenced by several explanatory variables (detailed above). The model relies on the cumulative normal distribution function, ensuring that all predicted probabilities remain within the [0,1] range. To ensure the robustness of the results, we have conducted all the estimations with robust standard errors. Moreover, in a standard probit regression, the estimated coefficients do not provide a direct interpretation of the magnitude of the effects but only indicate their sign and statistical significance. Therefore, marginal effects at means are computed, allowing complete coefficient interpretations. This method works for simple models, but models with interaction terms complicate the computation of marginal effects, especially with the large dataset. Thus, results will be interpreted using the sign of coefficients only for those models.

4 Results

4.1 Simple models

The first relevant result seen is the divergence between the two genders and the similarity of this result between anxiety and depression. Indeed, women are more likely to be anxious (depressed) than male, everything else equal we predict the average probability to be 7.2 (26.6) percentage points higher than male. This result aligns with the findings of Naima Z. Farhane-Medina (2022) and (Lowe et al. (2009)). Additionally, concerning the education level, we can observe a similar trend between the two disorders : as the level of education increases the likelihood of being anxious and depressed increases as well (see Figure 2). The effect is stronger for those with more than 12 years of education. While education is generally expected to protect against mental illnesses by providing better economic opportunities as demonstrated by Kondirolli and Sunder (2022) and Joannès et al. (2023), we have found the opposite effect. This positive relationship could be due to the fact that the higher education level is closely related to higher expectations of success, increased academic and professional pressures and a more competitive environment. Next, race is also a demographic variable that has an impact on mental condition. Interestingly, we have found the same results as those yielded by the "Black-White Mental Health Paradox" by Erving et al. (2019). Indeed, Black or African American (White) individuals have, everything else being equal, an average probability of 8.9 (1.2) percentage point lower (higher) than American Indian to be depressed (see Figure 3). This could be due to cultural differences in how depression is perceived and reported. For example, Rowden (2024) explains this result through self-esteem measures, how a person values and perceives themselves. He suggests that having a higher self-esteem can reduce the risk of developing symptoms of depression and that, on average, Black people in the United States have higher levels of self-esteem compared to White individuals. In addition, concerning the residential region we can observe that individuals living in the following regions : Midwest, South and West are more likely to suffer from anxiety and depression than individuals living in the Northeast region (see Figure 4). According to the Economic Policy Institute [8], the level of wages in the Midwest have the slowest growth rate of any other U.S. regions since the Great Recession and the COVID-19 pandemic. Despite its low rate of unemployment (3.2 % as of May 2023), the region had suffered from a rise of anti-workers policies in last 15 years that have deteriorated job quality and economic security for workers, such as restricting worker rights and preventing local governments from enhancing labor standards. Another explanation could be that regions like the Midwest are more rural than the Northeast, leading to reduced healthcare access and different cultural norms that can deteriorate mental well-being.

Surprisingly, we have predicted that for individuals who have already experienced homelessness, the average probability to be depressed (anxious) is 3.0 (6.6) percentage points lower than people that always lived in private residence everything else being equal. It contradicts the results of Moschion and van Ours (2022), that supports the idea that people that are homeless are more affected by mental disorders. However, our finding may be biased. Homeless individuals frequently encounter barriers to healthcare access, making them less likely to receive a formal diagnosis for mental health conditions which can result in an underestimation of anxiety and depression levels in the data. Once again, marital status affects anxiety and depression in the same way. Individuals who are married or have been married face a higher likelihood of depression compared to their never-married counterparts. For instance, we predicted, everything else being equal, that married individuals have an average probability of having depression (anxiety) of 25.5 (5.9) percent points higher than never married individuals. This result may seem counterintuitive as marriage is often associated with emotional support as shown by Walen and Lachman (2000). However, the increased responsibilities, marital conflicts, or stress from divorce or widowhood could contribute to higher depression rates. Conversely, being single may provide greater autonomy and less stress related to marital obligations. Finally the age, is the

[8]Economic Policy Institute. *Midwest Economic Recovery: Employment, Wages, and Disparities*, 2023

only variable in which we have observed an opposite effects between anxiety and depression (see Figure 1). Our results indicate that older individuals are less prone to anxiety. These following results align perfectly with article from Goodwin et al. (2020). During COVID-19, Varma et al. (2021) indicated that although older individuals were at higher risk from the virus, younger people were more affected by mental disorders due to stress and financial insecurity. Conversely for depression, we found a higher predicted probability for the oldest and younger age group. These results are in accordance with Zenebe et al. (2021) which mentions that older individuals are more affected. As people age, they may experience a reduction in social support due to retirement, the loss of friends or family members, leading to an increased risk of depression.

4.2 Interaction models

From the results of the interaction models, the effect of the employment status on the prevalence of the two disorder is relevant in this case study. Depression and anxiety follow the same trend. Indeed, having a part-time job contract or being unemployed implies a lower likelihood to be anxious or depressed than a full-time job contract. These following results goes with the findings of Virtanen et al. (2011). Indeed using data aggregating middle-aged British civil servants, they have found that working long hours is a predictor of depressive or anxiety symptoms. Long working hours can indicate the harmful effects of work exposure and stress. The interaction between employment status and education level shows that for individuals who are unemployed and have completed 12 years or more of education, the coefficient is positive compared to those with 0 to 8 years of education for anxiety and depression models (see Figure 5 and 6). This could be explained by the fact that individuals who have invested many years in their education and be unemployed may experience heightened stress or a sense of failure, leading to negative feelings that could easily escalate into depression. This result suggest that when considering both education level and unemployment, individuals with higher education are more likely to suffer from depression when unemployed. This contrasts with the regression of employment status alone, where being unemployed is typically associated with a lower probability of depression. Thus, the change can be explained by the fact that education influences how people perceive and experience unemployment.

5 Conclusion

To conclude, this study examined the demographic and socio-economic factors influencing the prevalence of anxiety and depression in the United States. The key finding challenges the common belief that individuals with lower socio-economic status are more prone to mental health issues. Instead, the research reveals that those with higher education levels and demanding jobs are more affected. This outcome aligns with the growing pressures and stress associated with modern, high-performance lifestyles, where individuals in active, high-pressure roles are more vulnerable. From an economic perspective, the global cost in lost productivity is closely tied to this key finding, since the most affected by anxiety and depression are the individuals who could potentially be the most productive, such as full-time workers. The analysis also highlights students as a vulnerable group, particularly in the U.S., where high tuition fees create additional stress. Early intervention is crucial to prevent further economic and social costs. Policy recommendations for the U.S. government include introducing regular mental health screenings across companies and universities to detect issues early and reduce their impact on employees and students. Additionally, increased funding for healthcare access in high-risk regions like the Midwest, as well as greater financial aid and scholarships for students, would alleviate financial pressures and, in turn, reduce stress and anxiety related to education costs. However, this research paper is based only on individuals diagnosed with a mental health disorder. Expanding the analysis to include a healthy control group would allow for more accurate comparisons. Additionally, the inclusion of the quantitative variable 'income' could have been highly important and determinant in understanding the relationship between socio-economic status and mental health.

Mental health remains a growing global issue, tied to modern life pressures. Addressing it requires structural changes in workplaces, educational systems, and government policies to avoid escalating social and economic costs.

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A Appendix

A.1 Detail description of the variables

A.1.1 Education

The school grade level is defined as :

- Current grade level for school-age children who attended school at any time in the past three months.
- Highest grade level completed for school-age children who have not attended school at any time within the past three months.
- Highest educational attainment for all adult clients, whether currently in school or not.

A.1.2 Age

Calculated from the client's date of birth at the midpoint of the state's elected reporting period.

A.1.3 Gender

Indicates the client's most recently reported sex.

A.1.4 Race

Specifies the client's most recently reported race :

- American Indian/Alaska Native : A person having origins in any of the original people of North America and South America (including Central America) and who maintains tribal affiliation or community attachment.
- Asian : A person having origins in any of the following people of the Far East, the Indian subcontinent, or Southeast Asia, including, for example : Cambodia, China, India, Japan, Korea, Malaysia, Pakistan, the Philippine Islands, Thailand and Vietnam.
- Black or African American : A person having origins in any of the Black racial groups of Africa.

- Native Hawaiian or Other Pacific Islander : A person having origins in any of the original peoples of Hawaii, Guam, Samoa or other Pacific Islands.
- White : A person having origins in any of the original people of Europe, the Middle East or North Africa.
- Some other race alone/two or more races combines :

  - Some other race alone : The client is not identified with any category above or whose origin group, because of area custom, is regarded as a racial class distinct from the above categories (do not use this category for clients indicating multiple or mixed races).

  - Two or more races : The state data system allows multiple race selection and more than one race is indicated.

A.1.5 Employment Status

Provides more detailed information on clients categorized as “not in the labor force” , meaning they are neither employed nor actively seeking work within the past 30 days.

- Full-time : Use state definition for full time employment; includes members of the Armed Forces, and clients in full-time Supported Employment.
- Part-time : Use state definition for part time; includes clients in part-time Supported Employment.
- Employed full-time/part-time not differentiated.
- Unemployed : Defined as actively looking for work or laid off from job (and awaiting to be recalled) in the past 30 days.
- Not in labor force : Defined as not employed and not actively looking for work during the past 30 days (i.e. people not interested in work or people who have been discouraged to look for work).

A.1.6 Residential Status

Specifies the client’s residential status at the time of discharge or the most recent available status.

- Experiencing homelessness : Person has no fixed address; includes homeless, shelters.
- Private residence : Includes independent living adults, dependent living adults (adult clients living in a house, apartment or other similar dwellings and are heavily dependent on others for daily living assistance) and children living in a private residence, regardless of living arrangement.
- Other : Clients living in a foster home or a care center, residential care facility, crisis residence, institutional care facility, jail and/or correctional facility.

A.1.7 Marital Status

Identifies the client's marital status :

- Never married : Includes clients who are single or whose only marriage was annulled.
- Now married : Includes married couples, those living together as married, living with partners or cohabitating.
- Separated : Includes those legally separated or otherwise absent from spouse because of marital discord.
- Divorced, widowed.

A.1.8 Region

Indicates the client's residential region based on the divisions used by the U.S. Census Bureau.

- Other jurisdictions : Puerto Rico, Republic of Palau, and the Commonwealth of the Northern Mariana Islands.
- Northeast : New England Division (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont) and Middle Atlantic Division (New Jersey, New York, Pennsylvania).
- Midwest : East North Central Division (Illinois, Indiana, Michigan, Ohio, Wisconsin) and West North Central Division (Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota).

- South : South Atlantic Division (Delaware, District of Columbia, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, West Virginia), East South Central Division (Alabama, Kentucky, Mississippi, Tennessee) and West South Central Division (Arkansas, Louisiana, Oklahoma, Texas).
- West : Mountain Division (Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, Wyoming) and Pacific Division (Alaska, California, Hawaii, Oregon, Washington).

A.1.9 Not Labor Force Status

Provides more detailed information on clients categorized as "not in the labor force" , meaning they are neither employed nor actively seeking work within the past 30 days.

A.2 Summary Statistics of the Variables

Table 1: Distribution of Variables

Variables	Levels	Headcount	Frequency (in %)
Education	Special education	170,518	1.09
	0 to 8 years	3,995,250	25.60
	9 to 11 years	2,877,855	18.44
	12 (or GED)	5,677,765	36.38
	More than 12 years	2,884,387	18.48
Age	**0 - 11 years**	1,634,582	10.47
	12 - 14 years	1,052,935	6.75
	15 - 17 years	1,084,875	6.95
	18 - 20 years	709,672	4.55
	21 - 24 years	896,853	5.75
	25 - 29 years	1,370,661	8.78
	30 - 34 years	1,393,604	8.93
	35 - 39 years	1,291,842	8.28
	40 - 44 years	1,112,034	7.13
	45 - 49 years	1,046,895	6.71
	50 - 54 years	1,097,638	7.03
	55 - 59 years	1,098,309	7.04
	60 - 64 years	829,484	5.32
	65 years and older	984,829	6.31
	Missing/unknown/not collected/invalid	1,562	0.01
Gender	**Male**	7,388,394	47.34
	Female	8,204,261	52.57
	Missing/unknown/not collected/invalid	13,120	0.08
Race	**American Indian / Alaska Native**	251,116	1.61
	Asian	170,121	1.09
	Black or African American	3,212,198	20.58
	Native Hawaiian or Other Pacific Islander	30,718	0.2
	White	9,819,863	62.92
	Some other race alone / two or more races	1,278,931	8.2
	Missing/unknown/not collected/invalid	842,828	5.4
Employment Status	Full-time	1,299,741	8.33
	Part-time	770,180	4.94

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Variables	Levels	Headcount	Frequency (in %)
	Employed full-time/part-time not differentiated	369,643	2.37
	Unemployed	2,760,056	17.69
	Not in labor force	4,783,314	30.65
	Missing/unknown/not collected/invalid	5,622,841	36.05
Residential Status	Experiencing homelessness	569,576	3.65
	Private residence	13,017,337	83.41
	Other	1,200,543	7.69
	Missing/unknown/not collected/invalid	818,319	5.24
Marital Status	**Never married**	8,771,997	56.21
	Now married	1,438,279	9.22
	Separated	550,188	3.53
	Divorced, widowed	1,716,405	11
	Missing/unknown/not collected/invalid	3,128,906	20.05
Region	Other jurisdictions	19,396	0.12
	Northeast	2,111,787	13.53
	Midwest	4,144,058	26.55
	South	7,309,936	46.84
	West	2,020,598	12.95
Not Labor Force Status	**Retired, disabled**	2,339,452	14.99
	Student	637,060	4.08
	Homemaker	163,235	1.05
	Sheltered/in competitive employment	30,951	0.2
	Other	1,612,616	10.33
	Missing/unknown/not collected/invalid	10,822,461	69.35
Anxiety	Disorder not reported	11,911,800	76.33
	Disorder reported	3,693,975	23.67
Depression	Disorder not reported	10,777,680	69.06
	Disorder reported	4,828,095	30.94

A.3 Econometrics Results and Graphics

Table 2: Results of the Simple Probit Model on Anxiety

	Coefficient	Standard Error
Age		
15-17 years (ref)	0	(.)
18-20 years	0.00395	(0.00405)
21-24 years	-0.0180^{***}	(0.00498)
25-29 years	-0.0383^{***}	(0.00499)
30-34 years	-0.0530^{***}	(0.00507)
35-39 years	-0.0594^{***}	(0.00511)
40-44 years	-0.0629^{***}	(0.00518)
45-49 years	-0.0567^{***}	(0.00517)
50-54 years	-0.0765^{***}	(0.00511)
55-59 years	-0.106^{***}	(0.00510)
60-64 years	-0.137^{***}	(0.00528)
65+ years	-0.212^{***}	(0.00531)
Education		
0-8 years (ref)	0	(.)
Special education	-0.237^{***}	(0.00613)
9-11 years	0.0692^{***}	(0.00304)
12 years (GED)	0.0852^{***}	(0.00291)
More than 12 years	0.142^{***}	(0.00317)
Race		
American Indian/Alaska Native (ref)	0	(.)
Asian	-0.153^{***}	(0.0110)
Black or African American	-0.218^{***}	(0.00703)
Native Hawaiian or Other Pacific Islander	-0.0471^{**}	(0.0183)
White	0.189^{***}	(0.00681)
Some other race alone/two or more races	0.0824^{***}	(0.00729)
Gender		
Male (ref)	0	(.)
Female	0.248^{***}	(0.00164)
Marital Status		
Never married (ref)	0	(.)
Now married	0.195^{***}	(0.00250)
Separated	0.123^{***}	(0.00354)
Divorced, widowed	0.122^{***}	(0.00227)
Not in Labor Force Status		
Retired, disabled (ref)	0	(.)
Student	0.0276^{***}	(0.00379)
Homemaker	0.0594^{***}	(0.00383)
Sheltered employment	-0.121^{***}	(0.00941)
Other	0.0679^{***}	(0.00198)

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	Coefficient	Standard Error
Residential Status		
Private residence (ref)	0	(.)
Experiencing Homelessness	-0.238^{***}	(0.00432)
Other	-0.293^{***}	(0.00286)
Region		
Northeast (ref)	0	(.)
Midwest	0.278^{***}	(0.00393)
South	0.0939^{***}	(0.00377)
West	0.264^{***}	(0.00412)
Other jurisdictions	-0.441^{***}	(0.0209)
Constant	-1.182^{***}	(0.00918)
Pseudo R2	0.0400	
Observations	3,307,031	

$^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 3: Average Marginal Effects of the Simple Model on Anxiety

	Coefficient	Standard Error
Age		
15-17 years (ref)	0	(.)
18-20 years	0.0012241	(0.0012566)
21-24 years	−0.0055383***	(0.0015327)
25-29 years	−0.0117238***	(0.0015308)
30-34 years	−0.0161245***	(0.0015520)
35-39 years	−0.0180495***	(0.0015598)
40-44 years	−0.0190741***	(0.0015800)
45-49 years	−0.0172396***	(0.0015797)
50-54 years	−0.0231117***	(0.0015561)
55-59 years	−0.0316947***	(0.0015440)
60-64 years	−0.0406108***	(0.0015821)
65+ years	−0.0609828***	(0.0015625)
Education		
0-8 years (ref)	0	(.)
Special education	−0.0600919***	(0.0014568)
9-11 years	0.0198407***	(0.0008632)
12 years (GED)	0.0245793***	(0.0008219)
More than 12 years	0.0417764***	(0.0009147)
Race		
American Indian/Alaska Native (ref)	0	(.)
Asian	−0.0400681***	(0.0028247)
Black or African American	−0.055342***	(0.0019343)
Native Hawaiian or Other Pacific Islander	−0.0128622**	(0.0049184)
White	0.0566042***	(0.0019000)
Some other race alone/two or more races	0.0236847***	(0.0020504)
Gender		
Male (ref)	0	(.)
Female	0.0724992***	(0.0004744)
Marital Status		
Never married (ref)	0	(.)
Now married	0.0590912***	(0.0007822)
Separated	0.0362876***	(0.0010816)
Divorced, widowed	0.0360587***	(0.0006846)
Not in Labor Force Status		
Retired, disabled (ref)	0	(.)
Student	0.0080788***	(0.0011177)
Homemaker	0.0175848***	(0.0011526)
Sheltered employment	−0.0336122***	(0.0024935)
Other	0.0201742***	(0.0005904)
Residential Status		
Private residence (ref)	0	(.)
Experiencing Homelessness	−0.0663412***	(0.0011011)
Other	−0.0799252***	(0.0007065)
Region		
Northeast (ref)	0	(.)
Midwest	0.0808413***	(0.0010693)
South	0.0255321***	(0.0009931)
West	0.0765188***	(0.0011351)
Other jurisdictions	−0.0938843***	(0.0035138)

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	Coefficient	Standard Error
Constant	-1.181536^{***}	(0.0091803)
Pseudo R2	0.0400	
Observations	3,307,031	

$^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 4: Results of the Probit Model with Interaction on Anxiety

	Coefficient	Standard Error
Employment Status		
Full-time (ref)	−0.0046	(0.0491)
Part-time	−0.0713	(0.0478)
Unemployed	−0.6224***	(0.0468)
Not in labor force	−0.2355***	(0.0460)
Employment Status # Region		
Full-time#Other jurisdictions	−0.8648***	(0.0625)
Full-time#Midwest	0.5738***	(0.0070)
Full-time#South	−0.0351***	(0.0070)
Full-time#West	0.1834***	(0.0073)
Part-time#Other jurisdictions	−0.3996***	(0.0749)
Part-time#Midwest	0.3684***	(0.0075)
Part-time#South	0.0153**	(0.0074)
Part-time#West	0.2411***	(0.0076)
Unemployed#Other jurisdictions	0.0566	(0.0350)
Unemployed#Midwest	0.6473***	(0.0051)
Unemployed#South	0.3498***	(0.0049)
Unemployed#West	0.5409***	(0.0052)
Not in labor force#Other jurisdictions	−0.4712***	(0.0194)
Not in labor force#Midwest	0.5796***	(0.0034)
Not in labor force#South	0.1408***	(0.0034)
Not in labor force#West	0.3058***	(0.0037)
Employment Status # Education		
Full-time#Special education	−0.2328***	(0.0288)
Full-time#9 to 11	−0.1129***	(0.0064)
Full-time#12 (or GED)	−0.0541***	(0.0055)
Full-time#More than 12	−0.0079	(0.0056)
Part-time#Special education	−0.3218***	(0.0173)
Part-time#9 to 11	−0.0129	(0.0089)
Part-time#12 (or GED)	0.0335***	(0.0082)
Part-time#More than 12	0.1149***	(0.0083)
Unemployed#Special education	−0.2634***	(0.0088)
Unemployed#9 to 11	−0.0040	(0.0041)
Unemployed#12 (or GED)	0.0431***	(0.0038)
Unemployed#More than 12	0.1488***	(0.0040)
Not in labor force#Special education	−0.6725***	(0.0053)
Not in labor force#9 to 11	−0.1453***	(0.0022)
Not in labor force#12 (or GED)	−0.0949***	(0.0020)
Not in labor force#More than 12	−0.0329***	(0.0023)
Employment Status # Gender		
Full-time#Female	0.2661***	(0.0024)
Part-time#Female	0.2435***	(0.0032)
Unemployed#Female	0.2239***	(0.0017)
Not in labor force#Female	0.2999***	(0.0013)
Age # Employment Status		
Full-time#18-20 years	−0.0308	(0.0165)
Full-time#21-24 years	−0.0229	(0.0160)
Full-time#25-29 years	−0.0046	(0.0159)
Full-time#30-34 years	−0.0035	(0.0159)
Full-time#35-39 years	−0.0263	(0.0159)
Full-time#40-44 years	−0.0398**	(0.0160)
Full-time#45-49 years	−0.0782***	(0.0161)
Full-time#50-54 years	−0.1043***	(0.0163)

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	Coefficient	Standard Error
Full-time#55-59 years	-0.1252^{***}	(0.0165)
Full-time#60-64 years	-0.1463^{***}	(0.0171)
Full-time#65+ years	-0.2620^{***}	(0.0186)
Part-time#18-20 years	-0.0132	(0.0100)
Part-time#21-24 years	-0.0682^{***}	(0.0098)
Part-time#25-29 years	-0.1014^{***}	(0.0096)
Part-time#30-34 years	-0.1239^{***}	(0.0097)
Part-time#35-39 years	-0.1312^{***}	(0.0099)
Part-time#40-44 years	-0.1629^{***}	(0.0102)
Part-time#45-49 years	-0.1748^{***}	(0.0105)
Part-time#50-54 years	-0.2076^{***}	(0.0106)
Part-time#55-59 years	-0.2310^{***}	(0.0109)
Part-time#60-64 years	-0.2619^{***}	(0.0119)
Part-time#65+ years	-0.3300^{***}	(0.0139)
Unemployed#18-20 years	0.05096^{***}	(0.00842)
Unemployed#21-24 years	0.01362	(0.00806)
Unemployed#25-29 years	0.00618	(0.00789)
Unemployed#30-34 years	0.01493	(0.00789)
Unemployed#35-39 years	0.00475	(0.00793)
Unemployed#40-44 years	0.01004	(0.00800)
Unemployed#45-49 years	-0.00180	(0.00806)
Unemployed#50-54 years	-0.04085^{***}	(0.00809)
Unemployed#55-59 years	-0.08631^{***}	(0.00819)
Unemployed#60-64 years	-0.1431^{***}	(0.00864)
Unemployed#65+ years	-0.2961^{***}	(0.00947)
Not in labor force#18-20 years	0.0538^{***}	(0.0032)
Not in labor force#21-24 years	0.0554^{***}	(0.0034)
Not in labor force#25-29 years	0.0465^{***}	(0.0031)
Not in labor force#30-34 years	0.0154^{***}	(0.0031)
Not in labor force#35-39 years	0.00169	(0.0031)
Not in labor force#40-44 years	-0.00909^{**}	(0.0031)
Not in labor force#45-49 years	-0.0151^{***}	(0.0031)
Not in labor force#50-54 years	-0.0530^{***}	(0.0030)
Not in labor force#55-59 years	-0.0949^{***}	(0.0030)
Not in labor force#60-64 years	-0.1302^{***}	(0.0031)
Not in labor force#65+ years	-0.2281^{***}	(0.0031)
Constant	-0.7169^{***}	(0.0458)
Pseudo R2	0.0381	
Observations	9,976,239	

$^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 5: Results of the Simple Probit Regression Model on Depression

	Coefficient	Standard Error
Age		
15–17 years (ref)	0	(.)
18–20 years	−0.01150***	(0.00379)
21–24 years	−0.12080***	(0.00470)
25–29 years	−0.19904***	(0.00473)
30–34 years	−0.23722***	(0.00481)
35–39 years	−0.23999***	(0.00484)
40–44 years	−0.20488***	(0.00490)
45–49 years	−0.12920***	(0.00487)
50–54 years	−0.05873***	(0.00479)
55–59 years	−0.02028***	(0.00477)
60–64 years	0.00778	(0.00492)
65 years and older	0.01109**	(0.00494)
Education		
0–8 years (ref)	0	(.)
Special education	−0.49893***	(0.00611)
9 to 11	0.07771***	(0.00280)
12 (or GED)	0.06302***	(0.00268)
More than 12	0.07533***	(0.00294)
Race		
American Indian/Alaska Native (ref)	0	(.)
Asian	−0.16467***	(0.01004)
Black or African American	−0.08919***	(0.00648)
Native Hawaiian or Other Pacific Islander	−0.01571	(0.01687)
White	0.01179*	(0.00631)
Some other race alone/two or more races	0.00895	(0.00678)
Gender		
Male (ref)	0	(.)
Female	0.26689***	(0.00153)
Marital Status		
Never married (ref)	0	(.)
Now married	0.25518***	(0.00236)
Separated	0.20904***	(0.00330)
Divorced, widowed	0.18306***	(0.00212)
Not in Labor Force Status		
Retired, disabled (ref)	0	(.)
Student	0.10510***	(0.00361)
Homemaker	0.16859***	(0.00369)
Sheltered/non-competitive employment	−0.12201***	(0.00920)
Other	0.17474***	(0.00185)
Residential Status		
Private residence (ref)	0	(.)
Experiencing Homelessness	−0.02988***	(0.00379)
Other	−0.33831***	(0.00265)
Region		
Northeast (ref)	0	(.)
Other jurisdictions	−0.28480***	(0.01708)
Midwest	0.24850***	(0.00365)
South	0.20506***	(0.00347)
West	0.17053***	(0.00385)

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	Coefficient	Standard Error
Constant	-0.87050^{***}	(0.00853)
Pseudo R2	0.0357	
Observations	3,307,031	

$^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 6: Average Marginal Effects of the Simple Model on Depression

	Coefficient	Standard Error
Age		
15–17 years (ref)	0	(.)
18–20 years	−0.01150***	(0.00379)
21–24 years	−0.12080***	(0.00470)
25–29 years	−0.19904***	(0.00473)
30–34 years	−0.23722***	(0.00481)
35–39 years	−0.23999***	(0.00484)
40–44 years	−0.20488***	(0.00490)
45–49 years	−0.12920***	(0.00487)
50–54 years	−0.05873***	(0.00479)
55–59 years	−0.02028***	(0.00477)
60–64 years	0.00778	(0.00492)
65 years and older	0.01109**	(0.00494)
Education		
0–8 years (ref)	0	(.)
Special education	−0.49893***	(0.00611)
9 to 11 years	0.07771***	(0.00280)
12 years (or GED)	0.06302***	(0.00268)
More than 12 years	0.07533***	(0.00294)
Race		
American Indian/Alaska Native (ref)	0	(.)
Asian	−0.16467***	(0.01004)
Black or African American	−0.08919***	(0.00648)
Native Hawaiian or Other Pacific Islander	−0.01571	(0.01687)
White	0.01179*	(0.00631)
Some other race alone/two or more races	0.00895	(0.00678)
Gender		
Male (ref)	0	(.)
Female	0.26689***	(0.00153)
Marital Status		
Never married (ref)	0	(.)
Now married	0.25518***	(0.00236)
Separated	0.20904***	(0.00330)
Divorced, widowed	0.18306***	(0.00212)
Not in Labor Force Status		
Retired, disabled (ref)	0	(.)
Student	0.10510***	(0.00361)
Homemaker	0.16859***	(0.00369)
Sheltered/non-competitive employment	−0.12201***	(0.00920)
Other	0.17474***	(0.00185)
Residential Status		
Private residence (ref)	0	(.)
Experiencing Homelessness	−0.02988***	(0.00379)
Other	−0.33831***	(0.00265)
Region		
Northeast (ref)	0	(.)
Other jurisdictions	−0.28480***	(0.01708)
Midwest	0.24850***	(0.00365)
South	0.20506***	(0.00347)
West	0.17053***	(0.00385)

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	Coefficient	Standard Error
Constant	-0.870497^{***}	(0.0085393)
Pseudo R2	0.0357	
Observations	3,307,031	

$^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 7: Results of the Probit Model with Interaction on Depression

	Coefficient	Standard Error
Employment Status		
Full-time	−0.09946**	(0.03346)
Part-time	−0.17083***	(0.03391)
Unemployed	−0.39144***	(0.03283)
Not in labor force	−0.28115***	(0.03262)
Employment Status # Education Interactions		
Full-time # Special education	−0.03191	(0.02663)
Full-time # 9 to 11 years	0.03217***	(0.00618)
Full-time # 12 (or GED)	0.07556***	(0.00534)
Full-time # More than 12	0.07723***	(0.00549)
Part-time # Special education	−0.42712***	(0.01648)
Part-time # 9 to 11 years	0.09010***	(0.00835)
Part-time # 12 (or GED)	0.06561***	(0.00769)
Part-time # More than 12	0.09430***	(0.00782)
Unemployed # Special education	−0.31175***	(0.00786)
Unemployed # 9 to 11 years	0.02036***	(0.00369)
Unemployed # 12 (or GED)	0.03725***	(0.00341)
Unemployed # More than 12	0.08632***	(0.00360)
Not in labor force # Special education	−0.78697***	(0.00537)
Not in labor force # 9 to 11 years	−0.01449***	(0.00204)
Not in labor force # 12 (or GED)	−0.01665***	(0.00186)
Not in labor force # More than 12	0.00253	(0.00213)
Employment Status # Gender Interactions		
Full-time # Female	0.24028***	(0.00228)
Part-time # Female	0.26521***	(0.00304)
Unemployed # Female	0.19758***	(0.00156)
Not in labor force # Female	0.31393***	(0.00121)
Employment Status # Region Interactions		
Full-time # Other jurisdictions	−0.83611***	(0.05265)
Full-time # Midwest	0.33154***	(0.00660)
Full-time # South	0.11042***	(0.00658)
Full-time # West	0.05164***	(0.00695)
Part-time # Other jurisdictions	−0.31003***	(0.06468)
Part-time # Midwest	0.28121***	(0.00701)
Part-time # South	0.19042***	(0.00685)
Part-time # West	0.07267***	(0.00718)
Unemployed # Other jurisdictions	0.12393***	(0.02891)
Unemployed # Midwest	0.53456***	(0.00428)
Unemployed # South	0.44329***	(0.00407)
Unemployed # West	0.31044***	(0.00447)
Not in labor force # Other jurisdictions	−0.28227***	(0.01557)
Not in labor force # Midwest	0.40507***	(0.00311)
Not in labor force # South	0.23657***	(0.00307)
Not in labor force # West	0.16731***	(0.00341)
Constant	−0.52058***	(0.03244)
Observations	9,976,338	
Pseudo R^2	0.0173	

$^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

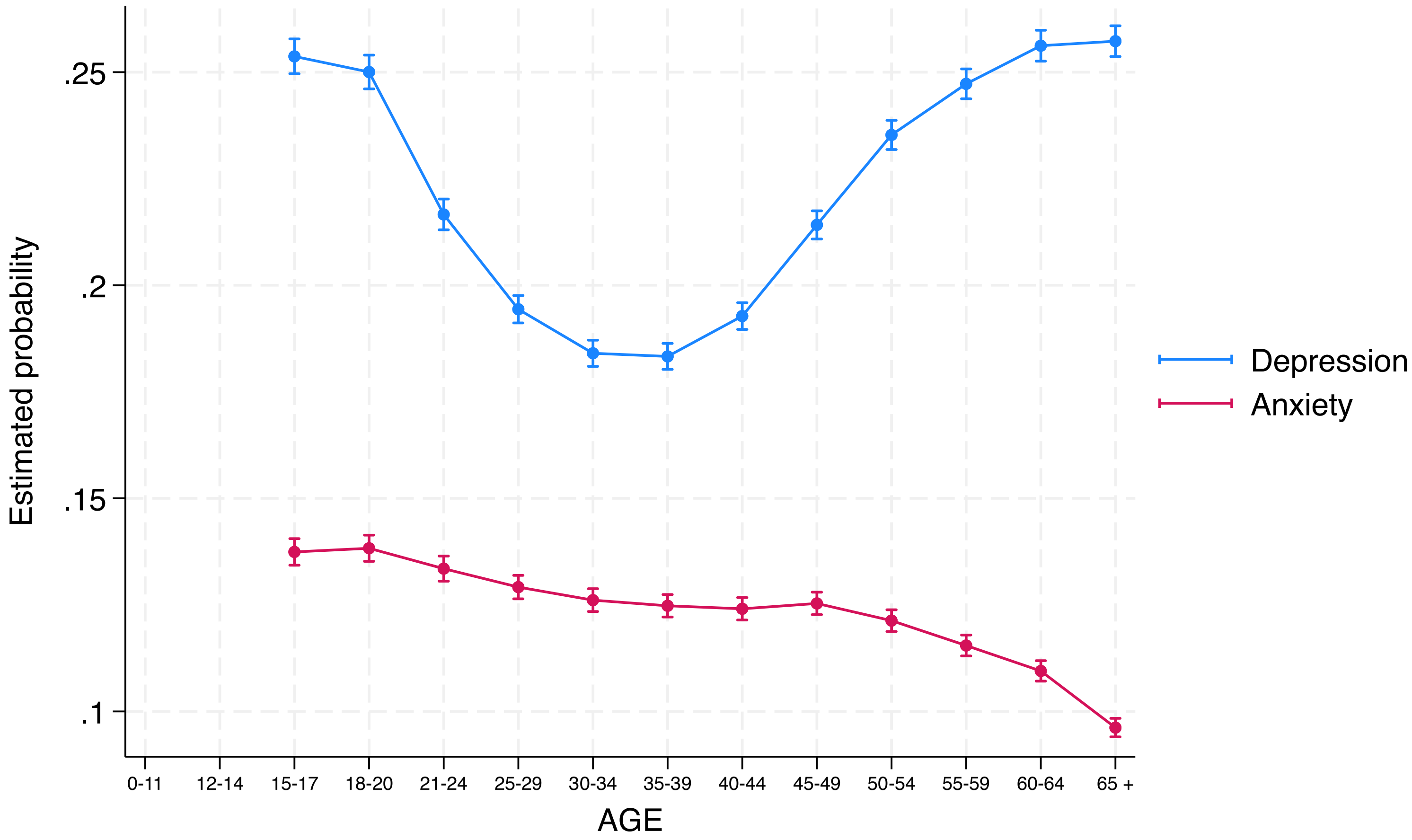

Figure 1: Predicted Probability of Being Anxious or Depressed by Age

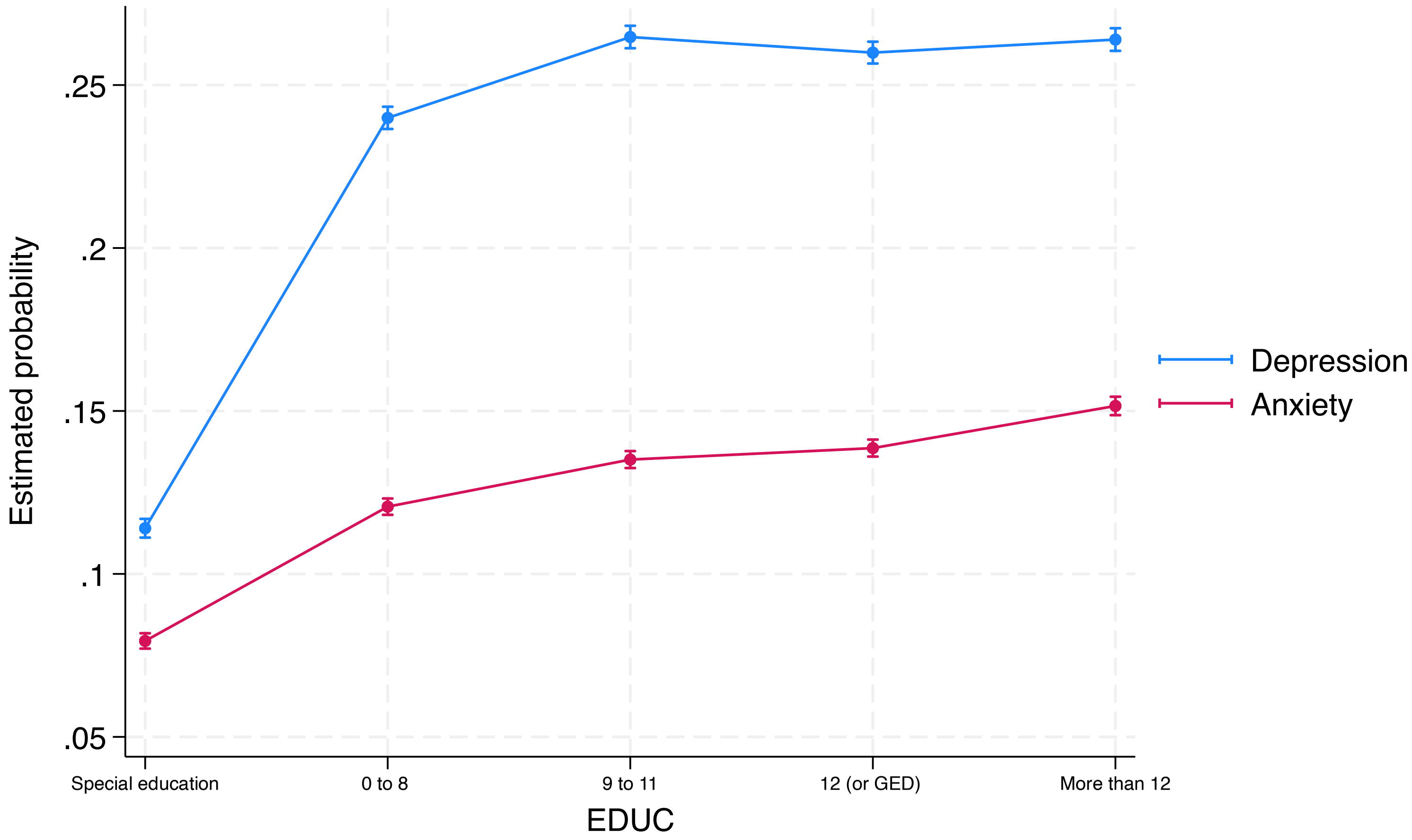

Figure 2: Predicted Probability of Being Anxious or Depressed by Education Level

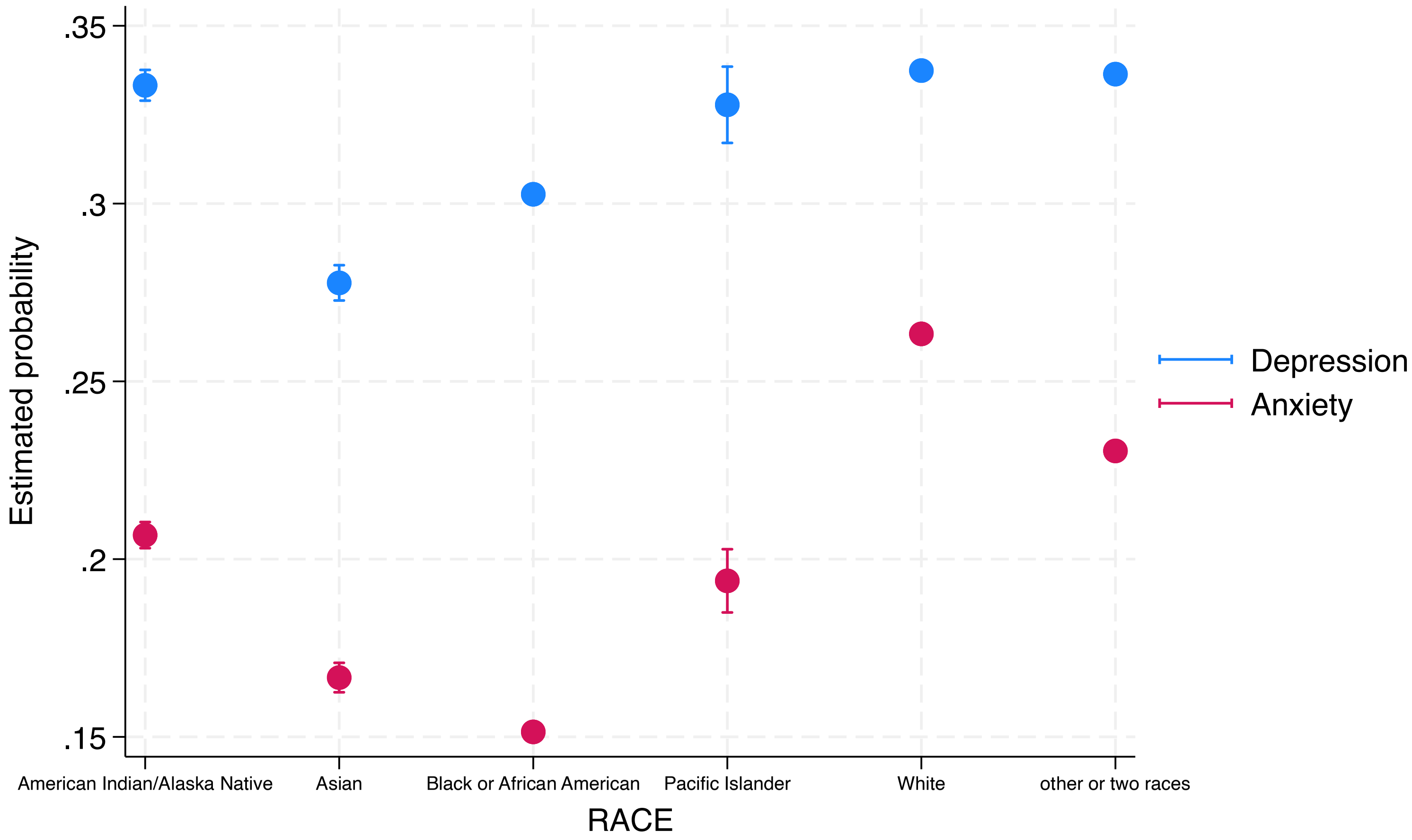

Figure 3: Predicted Probability of Being Anxious or Depressed by Race

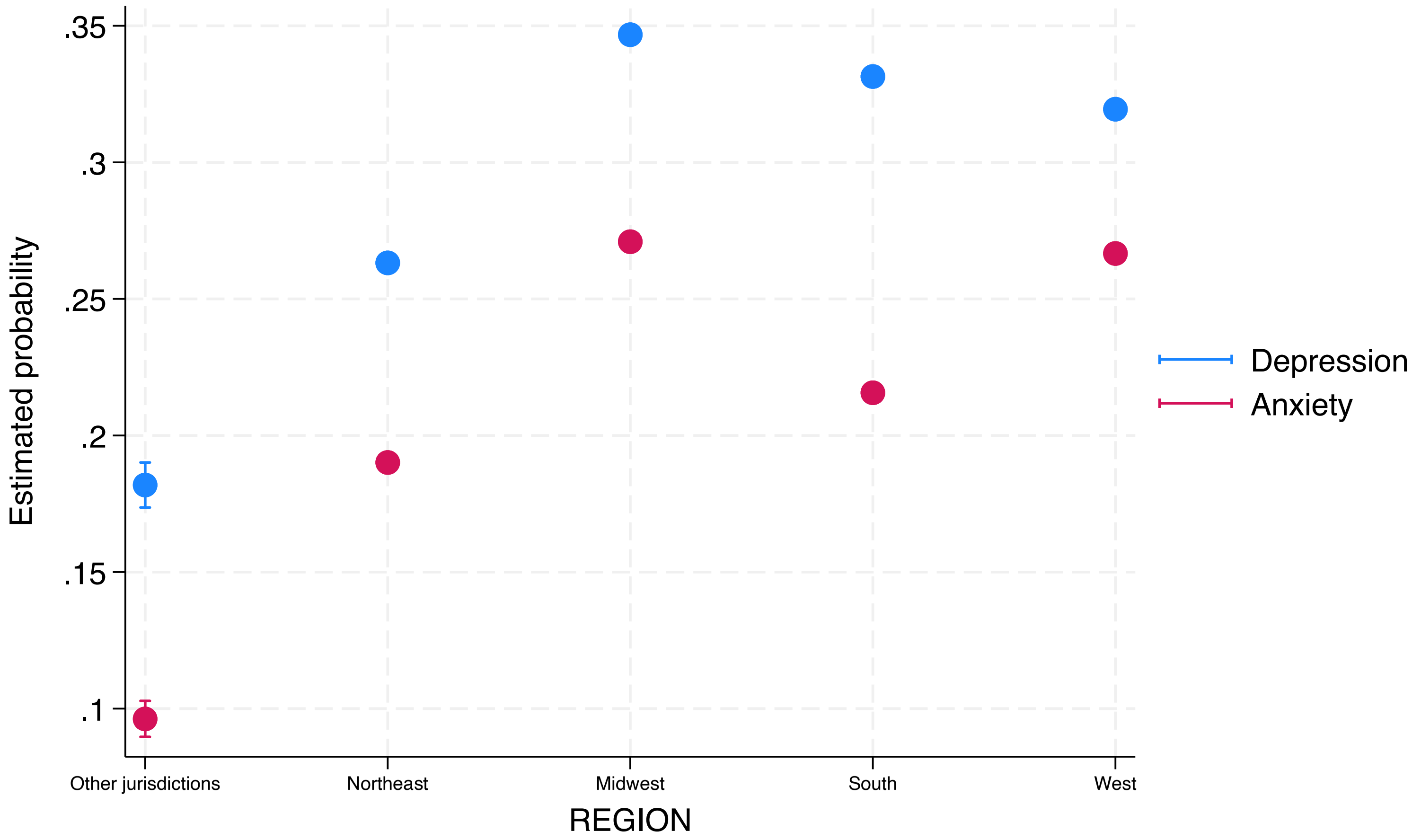

Figure 4: Predicted Probability of Being Anxious or Depressed by Region

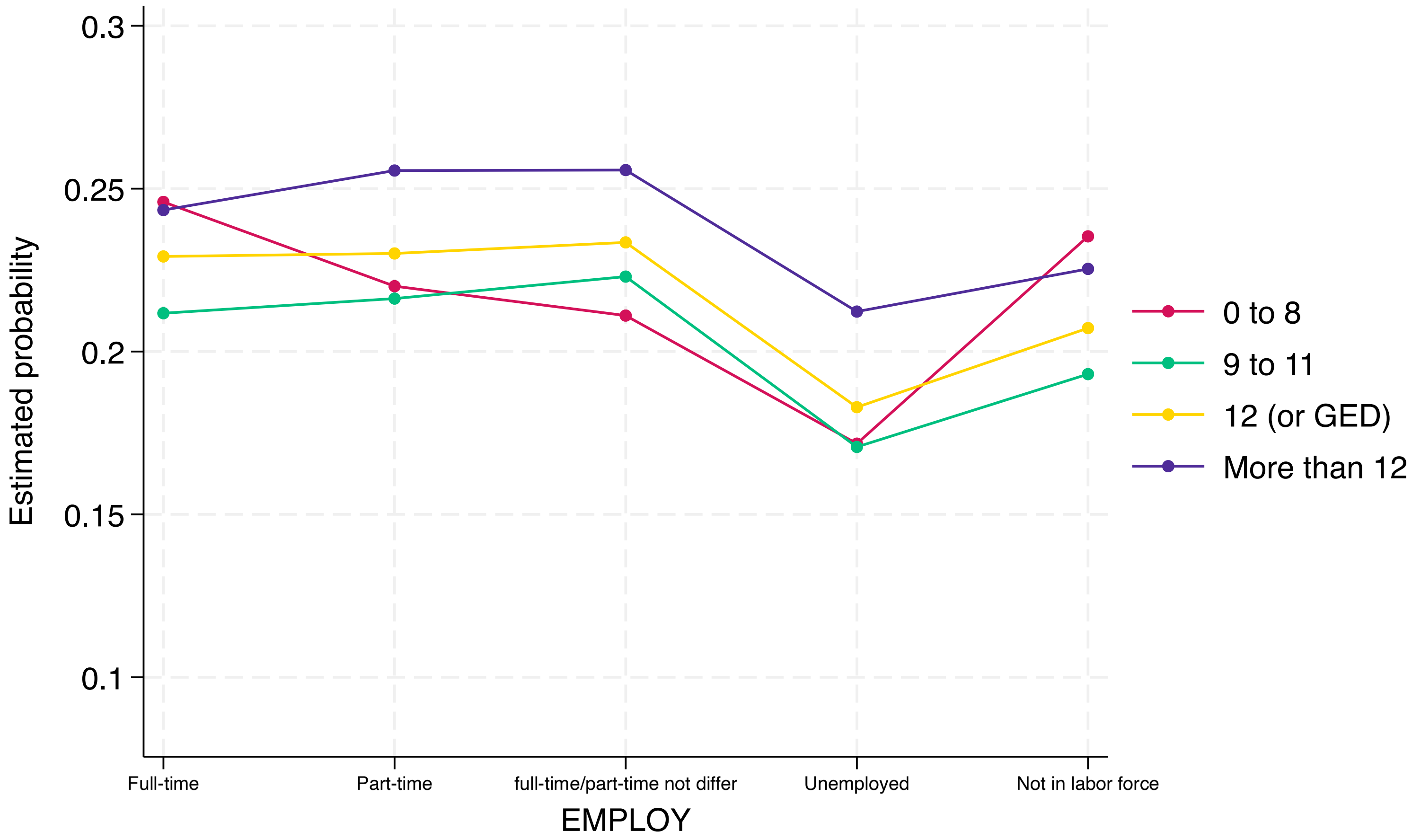

Figure 5: Predicted Probability of Being Depressed by Employment Status and Education Level

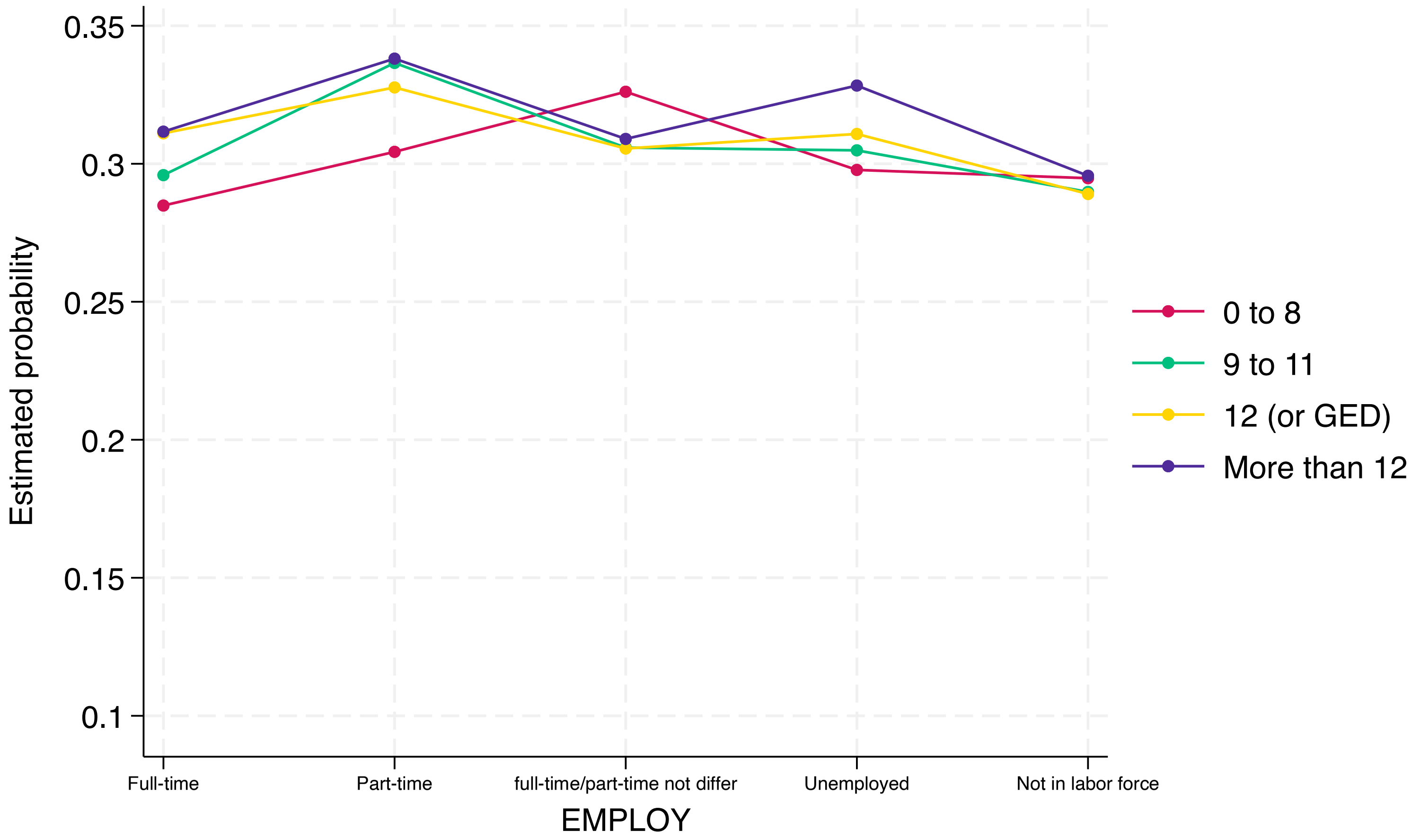

Figure 6: Predicted Probability of Being Anxious by Employment Status and Education Level